

Mingxuan Song

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Education

University College London

September 2024 – June 2028

MEng Robotics and AI

Major in Computer Science and Engineering

Related Coursework:

- Mathematics 1
- Mathematics 2
- Mechatronics
- Mechanical System
- Programming
- Electronics
- Control Theory 1
- Algorithms

St. Michael's school, Llanelli

September 2022 – June 2024

Alevel

Courses:

- Biology A*
- Physics A*
- Computer Science A*
- Mathematics A*
- Further Mathematics A

St. Edmund's College, Ware

September 2020 – June 2022

GCSE

Courses:

- Chemistry 9
- Biology 9
- Physics 9
- Computer Science 9
- Mathematics 9
- Design and Technology 9
- Chinese 9
- ICTs 8
- Religious Studies 7
- English as a Second Language 6

Projects

TianShu - a half Humanoid Robotic System with Mecanum Wheel Base

March 2025

Skills: Mechatronics, Electronics, Object Detection, Path Planing

- Created a Mecanum-wheel humanoid robot designed for industrial automation, specializing in object recognition, assembly, and defect detection.
- Featuring two 7-DOF robotic arms with independently controlled grippers.

OmniDexter - A Tendon Driven 4 DOF Wrist with Gripper

December 2024

Skills: Mechatronics

- Created 3-degree-of-freedom tendon-driven robotic wrist integrated with a 2-DOF underactuated gripper, powered by a single degree of actuation.

Text Generating Model

May 2024

Skills: Artificial Intelligence

- Created a text generation model in **Python** using PyTorch and the multi-head attention mechanism in the Transformer model.
- Trained to simulate Shakespeare's style.

Planning App

April 2024

Skills: Software Engineering

- Programmed a planning app in **Python** using OOP techniques, offering customizable yearly, monthly, and daily plans.
- Implemented modular interface to encourage wider range of customer to use.

Bipedal Robot

December 2023

Skills: Robotics, Design, Automation, Failure Analysis

- Developed and designed a robot with 3 Degrees of Freedom per leg using **Fusion 360** and **Arduino IDE**.
- Implemented inverse kinematics to optimize the robot's movement and analyzed mechanical failures to improve the overall design.

Medical Reminder Box

May 2022

Skills: Product Design, Automation, Electronics

- Developed a medical reminder box using **Fusion 360**, **Arduino IDE**, and **Python**, allowing users to set reminders for medication intake.
- Designed an automated system for dispensing medications with a child-lock mechanism for safety.
- The system also sends notifications to relatives if a dose is missed, in case of any emergency.

Doctor Octopus Robotic Arm

June 2021

Skills: Robotics, Mechanical Design, Automation

- Designed a 2 degrees of freedom tendon-driven robotic arm and a 6 degrees of freedom robotic claw using **Fusion 360** and **Arduino IDE**.
- Designed to be able to catch spherical or cylindrical objects.

Achievements

First in UCL First Year Robotic Challenge

June 2025

Second in AIESEC's Robotic Hackathon

March 2025

Presenter in UK 6th Robotic Manipulation Workshop

January 2025

Silver award in BPhO Online Senior Physics Challenge

January 2023

Merit award in Senior Kangaroo

December 2022

Gold award in UKMT Senior Mathematics Challenge

November 2022

Merit award in Pink Kangaroo

February 2022

Silver award in BPhO Intermediate Physics Challenge

January 2022

Gold award in UKMT Intermediate Mathematics Challenge

January 2022

Gold award in UKMT Senior Mathematics Challenge

December 2021

Extracurriculars

Humanoid Robot Research

Oct 2024 – Present

- Design and develop a PCB for controlling a 22 Degrees of Freedom humanoid robot.
- Focused on automation and performance improvements for the robot's real-time motion control.

AI Society Member

Sept 2024 – Present

- Conduct research in applying computer vision to medical imaging, help with diagnosis.

Kaggle competition

Aug 2024 – Present

- Participated in House Price and Titanic survival prediction challenges, focusing on data analysis, model optimization, and automation techniques.

Work Experience

UCL

Present

Role: Computer Science Ambassadors

- Conducting workshops on computing basics and introduced junior school students to robotics.

Technocamps, Swansea University

July 2023

Role: Workshop Assistant

- Conducted workshops on computing basics and introduced junior school students to robotics.
- Assisted in organizing a regional robotics competition and coordinating event logistics.
- Collaborated with team to provide project support, ensuring smooth competition operation and supporting students with problem-solving during failure scenarios.